CONFIDENTIAL: contains the reference answer and rubric. Never place this document where a model can read it.

dlbcl_0004 · v2 case pack

Large B-cell lymphoma with a late FISH addendum and an outdated R-CHOP plan: tumor-board plan

Difficulty: advanced · Chart exported 2026-09-21 · Review status: **PENDING_ONCOLOGIST** · Harbor tasks: dlbcl_0004_double_hit_addendum_v2_narrative, dlbcl_0004_double_hit_addendum_v2_fhir

Item	Summary
Patient	54-year-old man, ECOG 1. Right cervical nodes and a 12 cm anterior mediastinal mass narrowing the SVC; above the diaphragm only.
Core teaching point	FISH, not morphology or IHC, defines double-hit lymphoma.
Real-world twist	Pathology was signed out as DLBCL NOS and an R-CHOP plan was documented. A later FISH addendum shows MYC and BCL2 rearrangements; the problem list was never updated.
Key decisions	• Revise the diagnosis to HGBL with MYC and BCL2 rearrangements (dangerous if missed) • Cancel the outdated R-CHOP plan; DA-EPOCH-R x6 or a clinical trial • Bulky stage II, IPI 1, but double-hit biology dominates the risk • MRI brain and CSF; CNS prophylaxis considered; mandatory G-CSF; TLS measures; sperm banking
For the oncologist	• R-CHOP alone as the preferred regimen costs points but is not dangerous. Is that right for bulky stage II double-hit disease? • An extra must-not (dangerous) was added for dismissing CNS risk because of a low CNS-IPI. Keep it?

Synthetic training case for AI evaluation. Not clinical decision support. No real patient data.

A. Task instruction (model-visible)

````
You are the consulting hematologist-oncologist for this patient.

This is a synthetic training case for AI evaluation. It is not clinical decision support.

The chart export is in `/app/case/chart/`: a patient summary followed by dated notes, reports and
laboratory results, numbered in chronological order.

The patient will be presented at lymphoma tumor board tomorrow. Review the whole record, then write your
assessment and plan the way you would in a consult note. Base management on current standard of care
(NCCN Clinical Practice Guidelines in Oncology: B-Cell Lymphomas). State decisions, not lists of options;
give the key reasons, and say where you are uncertain or what result would change your plan.

Write your answer to:

```text
/app/final_answer.json
```

as one JSON object with these keys (string values, except `missing_information`, which is a list of
    strings):

- `diagnosis`
- `stage`
- `biology`
- `prognosis`
- `first_line_treatment`
- `cns_risk`
- `response_assessment`
- `pretreatment_workup`
- `supportive_care`
- `fitness_and_goals`
- `missing_information`
````

B. Patient chart (model-visible, narrative rendering)

The FHIR rendering carries the same facts, generated from the same source file (cases_src/dlbcl_0004/case.toml). Files appear in chronological order, as the model sees them.

chart/00_patient_summary.txt

```
PATIENT SUMMARY (EHR export 2026-09-21)

Name: Daniel Sample (synthetic)    MRN: OW-0004
DOB: 1972-01-30 (54 y)    Sex: male

PROBLEM LIST
- Obstructive sleep apnea syndrome (since 2019). Moderate (AHI 22 on 2019 sleep study). Uses CPAP nightly
    with good adherence.
- Mild intermittent asthma (since 1990). Albuterol as needed, about once a month. No hospitalizations or
    oral steroids in over 10 years.
- Gastroesophageal reflux disease (since 2016). Controlled on omeprazole.
- Obesity (since 2015). BMI 30.2 (September 2026).

ACTIVE MEDICATIONS
- Omeprazole 20 mg capsule, 1 capsule by mouth daily before breakfast
- Albuterol HFA 90 mcg/actuation inhaler, 2 puffs inhaled every 4-6 hours as needed for wheeze
- Cetirizine 10 mg tablet, 1 tablet by mouth daily as needed for seasonal allergies

ALLERGIES
- Penicillin: Urticaria (childhood)

This is a synthetic training case for AI evaluation. It is not clinical decision support.
```

chart/01_2026-08-27_referral.txt

```
PRIMARY CARE REFERRAL TO HEMATOLOGY-ONCOLOGY
Date: 2026-08-27    Author: Synthetic Primary Care Physician

Reason for referral: rapidly enlarging right neck mass, abnormal neck ultrasound.

Mr. Sample is a 54-year-old accountant with obstructive sleep apnea on CPAP, mild intermittent asthma
and GERD. He noticed a right-sided neck lump about four weeks ago; it has roughly doubled in size since.
A 10-day course of azithromycin (penicillin allergy) for presumed reactive adenitis made no difference.
Over the last two weeks he has had mild difficulty swallowing solid food (meat and bread "stick"), which
he first put down to his reflux. He denies fevers, night sweats or weight loss.

Neck ultrasound 2026-08-25: multiple enlarged, rounded right level II-IV lymph nodes with loss of the
fatty hilum, largest 3.6 cm. Not cystic.
I have ordered contrast CT of the neck and chest. Please see urgently.
```

chart/02_2026-09-03_ct.txt

CT NECK AND CHEST WITH IV CONTRAST
Date: 2026-09-03 Author: Synthetic Radiologist

CLINICAL INDICATION: Enlarging right neck mass, dysphagia.

TECHNIQUE: Axial CT of the neck and chest after 80 mL IV iodinated contrast; multiplanar reformats.

FINDINGS:
Neck: conglomerate of enlarged right level II, III and IV lymph nodes, largest 3.8 x 3.2 cm (level III), with mild mass effect on the right lateral pharyngeal wall. Oropharyngeal and laryngeal airway patent. No left cervical adenopathy. Thyroid normal.
Chest: lobulated soft-tissue mass in the anterior mediastinum measuring 12.0 x 8.6 x 9.8 cm (AP x TR x CC), mildly heterogeneous, insinuating around the great vessels. The superior vena cava is narrowed by extrinsic compression to about 50% of its normal caliber over 3 cm; contrast opacifies the SVC and right atrium normally with no filling defect to suggest thrombus and no prominent chest-wall collaterals. The trachea is displaced slightly to the left without significant luminal narrowing (narrowest cross-sectional diameter 15 mm). No hilar or axillary adenopathy. No pericardial or pleural effusion. Lungs clear. Upper abdomen on included images unremarkable.

IMPRESSION:
1. Large anterior mediastinal mass (12.0 cm) with right cervical lymphadenopathy, most consistent with lymphoma.
2. Moderate extrinsic narrowing of the SVC without thrombosis. Airway patent.

chart/03_2026-09-04_labs.txt

LABORATORY RESULTS: Complete blood count with differential
Collected: 2026-09-04 Synthetic Clinical Laboratory

Test	Result	Flag	Units	Reference
Hemoglobin	12.6	L	g/dL	13.5–17.5
MCV	88		fL	80–100
WBC	8.1		x10^3/uL	4.0–11.0
Absolute neutrophils	5.6		x10^3/uL	1.8–7.7
Absolute lymphocytes	1.5		x10^3/uL	1.0–4.8
Platelets	225		x10^3/uL	150–400

chart/04_2026-09-04_labs.txt

LABORATORY RESULTS: Comprehensive metabolic panel and tumor markers
Collected: 2026-09-04 Synthetic Clinical Laboratory

Test	Result	Flag	Units	Reference
Sodium	139		mmol/L	135–145
Potassium	4.3		mmol/L	3.5–5.1
Chloride	102		mmol/L	98–107
CO2, total	25		mmol/L	22–29
BUN	14		mg/dL	7–20
Creatinine	0.9		mg/dL	0.7–1.3
eGFR (CKD-EPI 2021)	101		mL/min/1.73m2	>=60
Glucose	96		mg/dL	70–99
Calcium	9.5		mg/dL	8.6–10.3
Albumin	4.0		g/dL	3.5–5.0
Total protein	7.1		g/dL	6.0–8.3
AST	29		U/L	10–40
ALT	33		U/L	7–56
Alkaline phosphatase	94		U/L	40–129
Bilirubin, total	0.6		mg/dL	0.2–1.2
Uric acid	7.6	H	mg/dL	3.4–7.0
Phosphorus	3.9		mg/dL	2.5–4.5
LDH	740	H	U/L	135–225
Beta-2 microglobulin	2.6	H	mg/L	0.8–2.2

chart/05_2026-09-04_labs.txt

LABORATORY RESULTS: Infectious disease serologies
Collected: 2026-09-04 Synthetic Clinical Laboratory

Test	Result	Flag	Units	Reference
Hepatitis B surface antigen	Nonreactive			Nonreactive
Hepatitis B core antibody, total	Nonreactive			Nonreactive
Hepatitis B surface antibody	Reactive			Nonreactive
Hepatitis C antibody	Nonreactive			Nonreactive
HIV-1/2 Ag/Ab screen	Nonreactive			Nonreactive

chart/06_2026-09-08_consult.txt

HEMATOLOGY-ONCOLOGY NEW PATIENT CONSULTATION
Date: 2026-09-08 Author: Synthetic Hematologist-Oncologist

HPI: 54M with a rapidly enlarging right neck mass for about five weeks and a large anterior mediastinal

mass on CT 2026-09-03. Mild dysphagia to solids (no odynophagia; liquids fine). For about ten days he has noticed mild facial fullness on waking that resolves within an hour of getting up. Mild dyspnea climbing two flights of stairs; none at rest. No orthopnea, stridor, hoarseness, cough or chest pain. No fevers, night sweats or weight loss (weight stable at 98 kg). No headache, visual change, confusion, focal weakness or numbness. No testicular swelling.

PMH: obstructive sleep apnea on CPAP (2019), mild intermittent asthma (albuterol about monthly), GERD, obesity. PSH: laparoscopic appendectomy (2004). No prior cancer, chemotherapy or radiation. No known heart disease.
Medications: see medication list.
Allergies: penicillin (urticaria in childhood).
Social: works full time as an accountant. Married, lives with his wife (46) and two children, ages 9 and 12. Never smoker. Alcohol: 2-3 beers per week. Lives 20 minutes from the cancer center.
Family history: mother with breast cancer at 70; no hematologic malignancy.

Exam: T 36.8 C, HR 84, BP 134/82, RR 16, SpO2 97% RA. Weight 98 kg, height 180 cm (BMI 30.2, BSA 2.18 m2). Firm, fixed, non-tender right cervical nodal mass about 5 cm (levels II-IV). No left cervical, supraclavicular, axillary or inguinal lymphadenopathy. Mild facial fullness; no plethora, no distended neck or chest-wall veins, no stridor. Lungs clear. Abdomen soft, no hepatosplenomegaly. Testes normal without masses. Neuro: alert and oriented, cranial nerves intact, strength 5/5, sensation intact.

ECOG performance status 1: still working, but tires by mid-afternoon and has stopped going to the gym.

Plan today: labs reviewed (LDH, uric acid, serologies); FDG PET-CT; echocardiogram; ENT for excisional biopsy of a right cervical node (scheduled 2026-09-09). Discussed sleeping with the head of the bed raised and to call or present to the emergency department for worsening facial swelling, breathlessness, noisy breathing or trouble swallowing liquids.

chart/07_2026-09-10_pet_ct.txt

FDG PET-CT, SKULL BASE TO MID-THIGH
Date: 2026-09-10   Author: Synthetic Nuclear Medicine Physician

CLINICAL INDICATION: Cervical and anterior mediastinal masses, suspected lymphoma. Initial staging.

TECHNIQUE: 12.4 mCi 18F-FDG IV; uptake time 60 minutes; serum glucose at injection 98 mg/dL. Low-dose non-contrast CT for attenuation correction and localization. Mediastinal blood pool SUVmax 1.9; liver SUVmax 2.7.

FINDINGS:
Head and neck: physiologic brain uptake (PET is insensitive for intracranial disease). Intensely FDG-avid right level II-IV cervical lymph nodes, largest 3.9 cm, SUVmax 21.3. Post-biopsy changes in the right neck. No FDG-avid left cervical or supraclavicular nodes. Waldeyer ring without focal uptake.
Chest: anterior mediastinal mass, 12.0 cm in greatest dimension, SUVmax 27.8, with a few photopenic (likely necrotic) foci; extrinsic compression of the SVC as on the diagnostic CT. No FDG-avid hilar, axillary or internal mammary nodes. No pulmonary nodules. No pleural or pericardial effusion and no abnormal pleural, pericardial or chest-wall uptake.
Abdomen and pelvis: no FDG-avid lymph nodes. Liver, spleen (10.5 cm), kidneys and adrenal glands without focal uptake. No FDG-avid disease below the diaphragm.
Musculoskeletal: no focal osseous or marrow FDG uptake.
Testes: no abnormal uptake.

IMPRESSION:
1. Intensely FDG-avid right cervical lymphadenopathy and a 12.0 cm FDG-avid anterior mediastinal mass, consistent with lymphoma.
2. No FDG-avid disease below the diaphragm. No FDG-avid extranodal organ, osseous or marrow lesions.

chart/08_2026-09-11_echocardiogram.txt

TRANSTHORACIC ECHOCARDIOGRAM
Date: 2026-09-11   Author: Synthetic Cardiologist

Indication: baseline before possible anthracycline-containing chemotherapy; mediastinal mass.
Left ventricle normal in size with normal wall thickness. LVEF 60% (biplane Simpson's).
No regional wall-motion abnormality. Normal diastolic function. Global longitudinal strain -20%.
Right ventricle normal size and function. Valves without significant stenosis or regurgitation.
No pericardial effusion. Anterior mediastinal soft tissue noted adjacent to the right heart border, without chamber compression.

chart/09_2026-09-12_pathology.txt

SURGICAL PATHOLOGY: EXCISIONAL BIOPSY, RIGHT CERVICAL LYMPH NODE
Date: 2026-09-12   Author: Synthetic Hematopathologist
Status: amended 2026-09-19

SPECIMEN: Right cervical lymph node, level III, excisional biopsy (3.6 x 2.9 x 2.2 cm), collected 2026-09-09. Fresh tissue sent for flow cytometry; blocks A1-A4.

FINAL DIAGNOSIS (2026-09-12): DIFFUSE LARGE B-CELL LYMPHOMA, NOT OTHERWISE SPECIFIED. See addendum.

MICROSCOPIC: Effaced nodal architecture with diffuse sheets of medium-to-large atypical lymphoid cells, round to irregular nuclei, vesicular chromatin and small nucleoli. Numerous mitoses and apoptotic bodies with a focal starry-sky pattern; geographic necrosis.
IMMUNOHISTOCHEMISTRY (adequate controls):

```
  CD20       positive (diffuse, strong)
  PAX5       positive
  CD3        negative in tumor cells (scattered background T cells)
  CD5        negative
  CD10       positive (diffuse)
  BCL6       positive
  MUM1/IRF4  negative (<10% of tumor cells)
  BCL2       positive in approximately 90% of tumor cells (strong)
  MYC        positive in approximately 80% of tumor cells
  Cyclin D1  negative
  CD23       negative
  CD30       negative
  TdT        negative
  Ki-67      approximately 95%
  EBER ISH   negative

FLOW CYTOMETRY: CD19+, CD20+, CD10+ monotypic B cells with lambda light-chain restriction, large forward
scatter. CD5 negative. CD34 negative.

COMMENT: Germinal-center B-cell-like immunophenotype by the Hans algorithm. FISH for MYC, BCL2
and BCL6 rearrangements has been requested on block A2; results will be reported in an addendum.

ADDENDUM (2026-09-19): FISH, interphase, block A2, 200 nuclei scored per probe.
MYC break-apart, IGH::MYC dual-fusion, BCL2 break-apart and BCL6 break-apart probes.
  MYC (8q24):   rearrangement detected (break-apart signal in 78% of nuclei; IGH::MYC fusion
                detected, consistent with t(8;14)(q24;q32))
  BCL2 (18q21): rearrangement detected (break-apart signal in 85% of nuclei; IGH::BCL2 fusion
                detected, consistent with t(14;18)(q32;q21))
  BCL6 (3q27):  no rearrangement detected
These FISH results change the classification of this lymphoma. The final diagnosis is amended per the
FISH findings; the treating hematologist was informed by telephone on 2026-09-19.
```

chart/10_2026-09-14_progress_note.txt

```
HEMATOLOGY-ONCOLOGY FOLLOW-UP (RESULTS VISIT)
Date: 2026-09-14    Author: Synthetic Hematologist-Oncologist

Results visit with the patient and his wife.

Pathology (2026-09-12): diffuse large B-cell lymphoma, germinal-center type by IHC; MYC and BCL2 protein
expression both high; Ki-67 about 95%. FISH sent, result pending.
PET-CT 2026-09-10 and echo 2026-09-11 reviewed with the patient (see reports). Labs 2026-09-04 reviewed.

Symptoms: neck mass a little larger. Dysphagia to solids unchanged. Morning facial fullness persists,
resolving within an hour. No stridor, orthopnea or dyspnea at rest. No fevers, night sweats or weight
loss. Weight 98 kg. Working half days.

Exam: unchanged right cervical mass. No stridor. No distended neck veins. Lungs clear.

Assessment: DLBCL, GCB type, with cervical and mediastinal disease.

Plan:
1. R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, prednisone) every 21 days x 6 cycles,
   cycle 1 to start 2026-09-22. Consent discussed; patient agrees.
2. Port placement with interventional radiology 2026-09-18.
3. Chemotherapy teaching visit 2026-09-19.
4. Will follow up the pending FISH result.
5. Return 2026-09-22 for cycle 1.
```

chart/11_2026-09-18_labs.txt

```
LABORATORY RESULTS: Pre-procedure chemistry (day of port placement)
Collected: 2026-09-18    Synthetic Clinical Laboratory

Test                                  Result  Flag  Units           Reference
Potassium                                4.4        mmol/L          3.5-5.1
Creatinine                               0.9        mg/dL           0.7-1.3
eGFR (CKD-EPI 2021)                      101        mL/min/1.73m2   >=60
Calcium                                  9.4        mg/dL           8.6-10.3
Phosphorus                               4.0        mg/dL           2.5-4.5
Uric acid                                7.9  H     mg/dL           3.4-7.0
LDH                                      812  H     U/L             135-225
```

chart/12_2026-09-21_progress_note.txt

```
HEMATOLOGY-ONCOLOGY BRIEF NOTE (PRE-TUMOR BOARD)
Date: 2026-09-21    Author: Synthetic Hematologist-Oncologist

Pathology addendum (FISH) issued 2026-09-19; hematopathologist called the same day. Results discussed
with the patient by telephone today; he understands the plan may change and wants the treatment with the
best chance of cure. He hopes to keep working part time from home if he can.

Port placed 2026-09-18 by interventional radiology (left chest, via left internal jugular vein) without
complication; tip at the cavoatrial junction.
```

Telephone symptom check: neck mass slightly larger; dysphagia to solids unchanged; morning facial fullness unchanged. No stridor, orthopnea, dyspnea at rest, headache or new neurological symptoms.

Pre-procedure labs 2026-09-18 reviewed (see results).

Cycle 1 chemotherapy (scheduled 2026-09-22) is ON HOLD pending lymphoma tumor board review on 2026-09-22.

C. Reference answer (hidden from the model)

diagnosis

High-grade B-cell lymphoma with MYC and BCL2 rearrangements (double-hit lymphoma; WHO-HAEM5 name DLBCL/HGBL-MYC/BCL2), germinal-center B-cell (GCB) phenotype by the Hans algorithm (CD10+, BCL6+, MUM1-). The original 2026-09-12 sign-out of DLBCL, NOS is superseded by the 2026-09-19 FISH addendum (IGH::MYC and IGH::BCL2 fusions; BCL6 not rearranged). CD23-, CD30- and the cervical nodal biopsy site argue against primary mediastinal B-cell lymphoma. The EHR problem list still reads DLBCL and should be updated.

stage

Lugano (Ann Arbor) stage II bulky (IIA-X): FDG-avid right cervical nodes and a 12.0 cm anterior mediastinal mass, all above the diaphragm. No B symptoms. No extranodal, splenic, osseous, marrow, testicular or CNS involvement on PET-CT. A 12 cm mass is bulky, so this is managed as advanced-stage disease rather than limited-stage disease.

biology

Double-hit: FISH shows MYC rearrangement (IGH::MYC, t(8;14)) and BCL2 rearrangement (IGH::BCL2, t(14;18)); BCL6 is not rearranged, so it is not triple-hit. It is also double-expressor by IHC (MYC 80%, BCL2 90%), but the FISH-defined double-hit status is the key classification and drives treatment. Ki-67 about 95%, starry-sky pattern. GCB cell of origin by Hans.

prognosis

IPI 1 (low by score): only LDH is above normal (740 rising to 812 U/L, about 3.6x ULN); age 54, ECOG 1, stage II and no extranodal sites score 0. Age-adjusted IPI 1 and CNS-IPI 1. These scores underestimate the risk: the MYC and BCL2 rearrangements (double-hit biology) dominate prognosis, with poorer outcomes after standard therapy and a higher risk of CNS relapse.

first_line_treatment

The 2026-09-14 plan of R-CHOP x6 starting 2026-09-22 was written before the FISH addendum and is outdated: cancel it. Recommend curative-intent DA-EPOCH-R (dose-adjusted etoposide, prednisone, vincristine, cyclophosphamide, doxorubicin, rituximab) for 6 cycles, or enrollment in a clinical trial for MYC/BCL2-rearranged lymphoma if one is available; R-hyper-CVAD/MA or R-CODOX-M/R-IVAC are acceptable intensive alternatives at experienced centers. R-CHOP alone is not preferred for rearrangement-defined double-hit lymphoma. Start promptly (within about 1 week) given rapid growth and SVC compression. Consolidative involved-site radiotherapy (ISRT) to the bulky mediastinal site should be decided at multidisciplinary review, particularly for residual PET-positive disease at the end of treatment.

cns_risk

Double-hit biology carries a high risk of CNS involvement and relapse regardless of the CNS-IPI of 1 (no kidney, adrenal or other high-risk extranodal sites). Before cycle 1: contrast-enhanced MRI brain and lumbar puncture with CSF cytology and flow cytometry. CNS prophylaxis should be given: intrathecal methotrexate is commonly given with DA-EPOCH-R per protocol, and systemic high-dose methotrexate (for example 2 cycles at 3 g/m2 or more) is an alternative; renal function is normal (creatinine 0.9), so high-dose methotrexate is feasible if chosen. If MRI or CSF shows CNS disease, the plan changes to a CNS-directed regimen.

response_assessment

Interim FDG PET-CT after cycles 2-4 (commonly after cycle 2 or 4 of DA-EPOCH-R) to confirm response; progression should prompt re-biopsy and a change of therapy, otherwise complete all 6 cycles. End-of-treatment FDG PET-CT 6-8 weeks after cycle 6 (and before any radiotherapy), read with Lugano 2014 criteria and the Deauville 5-point scale; Deauville 1-3 is a complete metabolic response. Residual Deauville 4-5 uptake in the mediastinum should be biopsied if it would change management, or treated with ISRT after multidisciplinary review.

pretreatment_workup

Hepatitis B serology done: HBsAg and anti-HBc nonreactive (anti-HBs reactive from vaccination), so no HBV antiviral treatment is needed before rituximab; HCV and HIV nonreactive. Echo LVEF 60%, so anthracycline (doxorubicin in DA-EPOCH-R) is permitted. Renal and hepatic function normal. Bone marrow biopsy is not needed because PET-CT shows no marrow uptake. Outstanding before cycle 1: contrast MRI brain, lumbar puncture with CSF cytology and flow cytometry, sperm banking, G6PD level in case rasburicase is needed, and confirmation that the new port is functioning.

supportive_care

Tumor lysis syndrome (TLS) risk is intermediate: 12 cm mass, LDH about 3.6x ULN, uric acid 7.6 rising to 7.9 mg/dL, normal creatinine. Start allopurinol 300 mg daily 2-3 days before cycle 1 with oral/IV hydration and TLS labs (potassium, phosphate, calcium, uric acid, creatinine, LDH) every 12-24 hours for the first 72 hours; use rasburicase if uric acid keeps rising or creatinine increases (check G6PD first). G-CSF is mandatory with DA-EPOCH-R: filgrastim from day 6 until neutrophil recovery (or pegfilgrastim on day 6) every cycle, with twice-weekly CBCs to guide dose adjustment. DA-EPOCH-R is a 96-hour continuous infusion of vesicant drugs and needs central venous access: use the left chest port placed 2026-09-18, with an ambulatory infusion pump or inpatient administration. Offer sperm banking before cycle 1. Mediastinal mass with SVC narrowing: monitor closely during cycle 1 (consider an inpatient start), warn him to report worsening facial swelling, dyspnea, orthopnea, stridor or trouble swallowing liquids, and avoid sedation or general anesthesia without anesthetic review. PJP prevention per institutional DA-EPOCH-R practice, antiemetics, and a bowel regimen with vincristine. Continue CPAP for sleep apnea and albuterol as needed.

fitness_and_goals

Fit for intensive curative therapy: age 54, ECOG 1, LVEF 60%, normal renal and hepatic function, and no comorbidity that limits treatment (sleep apnea on CPAP, mild intermittent asthma and GERD do not). His goal is cure and he wants the most effective treatment; discuss the inpatient or pump-based infusion schedule, twice-weekly labs and time off work, since he hopes to keep working part time from home.

missing_information

- Contrast MRI brain and CSF cytology/flow cytometry results (CNS staging not yet done)
- Whether sperm banking has been offered and the patient's fertility wishes
- G6PD status in case rasburicase is needed
- Clinical trial availability for MYC/BCL2-rearranged lymphoma

D. Grading rubric (hidden from the model)

Rubric dlbcl_0004.v2.0. Must and should items total 100 points; pass threshold 0.75. A **dangerous** item fails the case (reward 0) when a must item is missed or a must-not item is committed. Points for must-not items are deductions.

ID	Kind	Severity	Pts	Criterion (what the judge reads)
dx_hgbl_double_hit	must	dangerous	10	The response revises the diagnosis to high-grade B-cell lymphoma with MYC and BCL2 rearrangements (double-hit; WHO-HAEM5 DLBCL/HGBL-MYC/BCL2 also accepted) based on the 2026-09-19 FISH addendum, rather than keeping the original DLBCL NOS sign-out.
mn_dx_dlbcl_nos	must_not	dangerous	0	The response keeps the final diagnosis as DLBCL NOS (or otherwise denies that this is double-hit / HGBL with MYC and BCL2 rearrangements).
dx_gcb_hans	should	minor	2	The response notes a germinal-center B-cell (GCB) phenotype by the Hans algorithm (CD10+, BCL6+, MUM1-).
tx_plan_outdated	must	important	10	The response recognizes that the 2026-09-14 plan (R-CHOP x6 starting 2026-09-22) was made before the FISH addendum, is outdated, and must be cancelled/changed.
stage_ii_bulky	must	important	8	The response stages the disease as Lugano stage II bulky: right cervical and mediastinal nodes above the diaphragm only, 12.0 cm mediastinal mass, no extranodal disease.
mn_stage_iii_iv	must_not	dangerous	0	The response assigns stage III or stage IV.
bio_double_expressor_secondary	should	minor	3	The response notes the lymphoma is also double-expressor by IHC (MYC 80%, BCL2 90%) but that the FISH-defined double-hit status is the key classification.
prog_ipi1_biology_dominates	should	important	6	The response calculates IPI 1 (only LDH above normal; age 54, ECOG 1, stage II, no extranodal sites) and states that double-hit biology, not the IPI, dominates the risk.
tx_intensive_curative	must	dangerous	12	The response recommends curative-intent intensive therapy appropriate for double-hit lymphoma: DA-EPOCH-R x6 (preferred) or a clinical trial; R-hyper-CVAD/MA or R-CODOX-M/R-IVAC are acceptable alternatives.
mn_rchop_preferred	must_not	important	6	The response recommends R-CHOP alone (including proceeding with the 2026-09-14 plan) as the preferred/initial regimen.
mn_non_curative	must_not	dangerous	0	The response recommends palliative-only treatment, best supportive care, or hospice as the plan.
tx_isrt_consolidation	should	minor	3	The response considers consolidative involved-site radiotherapy (ISRT) to the bulky mediastinal site, e.g. for residual PET-positive disease or per multidisciplinary review.
cns_staging_mri_csf	must	important	6	The response treats double-hit biology as high CNS risk irrespective of the CNS-IPI and recommends baseline contrast MRI brain and lumbar puncture with CSF cytology and flow cytometry.
cns_prophylaxis	should	important	6	The response recommends (or says it should be considered) CNS prophylaxis, e.g. intrathecal methotrexate commonly given with DA-EPOCH-R, or high-dose methotrexate.
mn_cns_dismissed	must_not	dangerous	0	The response dismisses CNS risk as low (e.g. on the basis of a low CNS-IPI) and states that CNS evaluation or CNS prophylaxis is not needed.
resp_pet_lugano	must	important	6	The response plans an end-of-treatment FDG PET-CT read with Lugano criteria / Deauville 5-point scale (interim PET after cycles 2-4 is commonly used with DA-EPOCH-R).
sup_gcsf	must	important	5	The response states G-CSF support is mandatory with DA-EPOCH-R (filgrastim or pegfilgrastim each cycle).
sup_tls	must	important	6	The response recognizes tumor lysis risk (12 cm mass, LDH ~3.6x ULN, uric acid 7.6-7.9) and orders a uric-acid-lowering agent (allopurinol, or rasburicase if uric acid high/rising), hydration and TLS laboratory monitoring.
sup_central_access_infusion	should	minor	3	The response notes DA-EPOCH-R is a 96-hour continuous infusion requiring central venous access (the port placed 2026-09-18 can be used).

ID	Kind	Severity	Pts	Criterion (what the judge reads)
sup_fertility	should	minor	3	The response offers sperm banking / fertility preservation before chemotherapy.
sup_svc_airway	should	minor	3	The response addresses the mediastinal mass with SVC narrowing: monitoring for SVC syndrome / airway compromise and patient education.
work_hbv_reviewed	should	minor	2	The response confirms hepatitis B serology was checked (HBsAg and anti-HBc nonreactive) before rituximab.
work_echo_reviewed	should	minor	1	The response notes the baseline LVEF of 60% permits anthracycline.
fit_intensive	must	important	5	The response judges the patient fit for intensive curative therapy (age 54, ECOG 1, LVEF 60%, normal organ function) in line with his goal of cure.

E. Grader self-tests (regex mode, recomputed now)

Each adversarial answer must produce its expected verdict. 01_reference is the gold answer.

Answer	Reward	Passed	Expected	Dangerous errors fired
01_reference	1.00	yes	pass	-
bad_keeps_dlbcl_rchop	0.00	no	fail	dx_hgbl_double_hit, mn_dx_dlbcl_nos, tx_intensive_curative
bad_no_cns_eval	0.00	no	fail	mn_cns_dismissed
bad_stage_iv	0.00	no	fail	mn_stage_iii_iv
good_paraphrase_bcl6_negative	1.00	yes	pass	-
v1_gold_answer	0.47	no	fail	-

F. Reviewer sign-off

Reviewer	Credentials	Date	Verdict	Changes requested